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Updated: October 16, 2026
Course: **Statistics**

1 Basic probability theory

Exercise 1.1. For each of the following experiments, describe the sample space.

(a) Toss a coin four times.

(b) Count the number of insect-damaged leaves on a plant.

(c) Measure the lifetime (in hours) of a particular brand of light bulb.

(d) Record the weights of 10-day-old rats.

(e) Observe the proportion of defectives in a shipments of electronic components.

Solution. The sample space S is the set of all possible outcomes.

(a) For each coin toss in the series we receive either Head (H) or Tail (T). The example of the realization of the series of four tosses: THHH, meaning 1st toss gives T and the remaining tosses give H. The sample space S is the set of all such combinations. The cardinality of this set is $2^4 = 16$: on each position (toss) we have either T or H (2 options) and we observe four positions.

(b) The number of leaves should be a nonnegative integer. Thus, $S = \{0, 1, 2, 3, \ldots\}$.

(c) The lifetime may be less than an hour or some integer number of hours plus the remaining fraction of an hour. Moreover, the lifetime cannot be negative. Thus, $S = (0, T]$ for some constant $T < \infty$.

(d) Consider grams. Then we may say $S = (0, \infty)$. We may also define some upper bound, for example K. Then $S = (0, K]$.

(e) Define n to be the number of items in the shipment. Since we need a proportion, the sample space is $S = \{\frac{0}{n}, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \ldots, \frac{n}{n}\}$.

Exercise 1.2. Verify the following identities:

(a) $A \setminus B = A \setminus (A \cap B) = A \cap B^c$

(b) $B = (B \cap A) \cup (B \cap A^c)$

(c) $B \setminus A = B \cap A^c$

(d) $A \cup B = A \cup (B \cap A^c)$

[1]The author thanks Arsenii Scherbov for providing materials for the first half of the course.

Solution. When working with set expressions, Venn diagrams might be useful. However, the illustration does not constitute a formal proof and thus is not sufficient as an answer. To prove that the expression is true, it is necessary to explicitly show both directions: if $x \in$ lhs (left-hand side) then $x \in$ rhs (right-hand side) *and* if $x \in$ rhs then $x \in$ lhs.

(a) $A \setminus B \iff x \in A$ and $x \notin B \iff x \in A$ and $x \notin A \cap B \iff x \in A \setminus (A \cap B)$. At the same time, $x \in A$ and $x \notin B \iff x \in A$ and $x \in B^c \iff x \in A \cap B^c$.

(b) By definition $A \cup A^c = S$. Then using the Distributive Law $(B \cap A) \cup (B \cap A^c) = B \cap (A \cup A^c) = B \cap S = B$.

(c) $B \setminus A \iff x \in B$ and $x \notin A \iff x \in B$ and $x \in A^c \iff x \in B \cap A^c$.

(d) $A \cup B = A \cup [(B \cap A) \cup (B \cap A^c)] = A \cup (B \cap A) \cup A \cup (B \cap A^c) = A \cup [A \cup (B \cap A^c)] = A \cup (B \cap A^c)$.

Exercise 1.3. For events A and B, find formulas for the probabilities of the following events in terms of the quantities $\mathbb{P}(A)$, $\mathbb{P}(B)$, and $\mathbb{P}(A \cap B)$:

(a) either A or B or both,

(b) either A or B but not both,

(c) at least one of A or B,

(d) at most one of A or B.

Solution.

(a) "either A or B or both" means $A \cup B$, so we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

This is because $A \cup B = A \cup (B \cap A^c)$ (see previous exercise), from it $\mathbb{P}(A \cup B) = \mathbb{P}(A \cup (B \cap A^c)) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. In the last expression, the second equality uses the fact that A and $B \cap A^c$ are disjoint; the last equality comes from rearranging $\mathbb{P}(B) = \mathbb{P}((B \cap A) \cup (B \cap A^c)) = \mathbb{P}(B \cap A) + \mathbb{P}(B \cap A^c)$. It is easier to draw the diagram and note the fact that we count the area of the intersection two times if we do not subtract $\mathbb{P}(A \cap B)$.

(b) "either A or B but not both" is $(A \cap B^c) \cup (B \cap A^c)$, so we have

$$\begin{aligned}\mathbb{P}((A \cap B^c) \cup (B \cap A^c)) &= \mathbb{P}(A \cap B^c) + \mathbb{P}(B \cap A^c) \\ &= [\mathbb{P}(A) - \mathbb{P}(A \cap B)] + [\mathbb{P}(B) - \mathbb{P}(B \cap A)] \\ &= \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B).\end{aligned}$$

The easy way is to draw a diagram and note that this probability is equal to $\mathbb{P}(A \cup B) - \mathbb{P}(A \cap B)$.

(c) "at least one of A or B" is $A \cup B$, so we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

(d) "at most one of A or B" is $(A \cap B)^c$, so we have

$$\mathbb{P}((A \cap B)^c) = 1 - \mathbb{P}(A \cap B).$$

Note that this event also includes the possibility of not A and not B.

Exercise 1.4. Consider two different setups:

(a) A fair dice is cast until a 6 appears. What is the probability that it must be cast more than five times?

(b) A pair of events A and B cannot be simultaneously mutually exclusive and independent. Prove that if $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$, then:

- if A and B are mutually exclusive, they cannot be independent,
- if A and B are independent, they cannot be mutually exclusive.

Solution.

(a) It must be cast more than five times if we do not observe the appearance of 6 on first five casts. We do not observe 6 on each cast with probability 5/6. Since casts of a die are independent the probability to not receive 6 in first five casts is $(5/6)^5 \approx 0.4$.

(b)
- Let A and B be mutually exclusive. Suppose, for the sake of contradiction, that A and B are independent, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. However, from the initial property $A \cap B = \emptyset$ and $\mathbb{P}(A \cap B) = 0$ (mutually exclusive). Since we are given events A and B with positive probabilities, we come to a contradiction $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) = 0$ with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. Thus, mutually exclusive events cannot be independent.
- Independence of A and B together with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$ gives $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) > 0$. It means that $A \cap B \neq \emptyset$. Therefore, A and B are not mutually exclusive by definition. Thus, independent events cannot be mutually exclusive.

Exercise 1.5. Two coins, one with $\mathbb{P}(\text{head}) = u$ and one with $\mathbb{P}(\text{head}) = w$, are to be tossed together independently. Let

$$p_0 = \mathbb{P}(0 \text{ heads occur}), \quad p_1 = \mathbb{P}(1 \text{ heads occur}), \quad p_2 = \mathbb{P}(2 \text{ heads occur}).$$

Can u and w be chosen such that $p_0 = p_1 = p_2$? Prove your answer.

Solution. Start with probabilities: $p_0 = (1-u)(1-w)$, that is, first is T and second is T; $p_1 = (1-u)w + u(1-w)$, that is, first is T and second is H or first is H and second is T; and $p_2 = uw$, that is, both are H. Equating these probabilities, we receive

$$p_0 = p_2 \Rightarrow u + w = 1, \quad p_1 = p_2 \Rightarrow uw = \frac{w+u}{3}.$$

These together imply

$$u(1-u) = \frac{1}{3}.$$

This equation has no solution in the real numbers (more specifically, the right-hand side must be $\leq \frac{1}{4}$ for the solution in real numbers to exist). Thus, we cannot choose legitimate u and w to satisfy the conditions.

Exercise 1.6. Consider telegraph signals "dot" and "dash" sent in the proportion 3:4, where erratic transmissions cause a dot to become a dash with probability 1/4 and a dash to become a dot with probability 1/3.

(a) If a dash is received, what is the probability that a dash has been sent? If a dot is received, what is the probability that a dot has been sent?

(b) Assuming independence between signals, if the message dot-dot was received, what is the probability distribution of the four possible messages that could have been sent?

Solution. Let "DA" mean "dash", "DO" mean "dot", "R" mean "received", "S" mean "sent". From dot/dash 3:4 proportion we can calculate unconditional probabilities to observe dots and dashes:

$$\mathbb{P}(\text{dot sent}) := \mathbb{P}(\text{DOS}) = \frac{3}{3+4} = \frac{3}{7}, \quad \mathbb{P}(\text{dash sent}) := \mathbb{P}(\text{DAS}) = \frac{4}{3+4} = \frac{4}{7}.$$

From probabilities of mistakes we deduce other important objects, such as

$$\mathbb{P}(\text{DOR}|\text{DOS}) = 1 - \frac{1}{4} = \frac{3}{4}, \quad \mathbb{P}(\text{DAR}|\text{DAS}) = 1 - \frac{1}{3} = \frac{2}{3},$$

and

$$\mathbb{P}(\text{DOR}|\text{DAS}) = \frac{1}{3}, \quad \mathbb{P}(\text{DAR}|\text{DOS}) = \frac{1}{4}.$$

(a) Using the Bayes' rule, we calculate the following:

$$\begin{aligned}\mathbb{P}(\text{DAS}|\text{DAR}) &= \frac{\mathbb{P}(\text{DAR}|\text{DAS}) \cdot \mathbb{P}(\text{DAS})}{\mathbb{P}(\text{DAR}|\text{DAS}) \cdot \mathbb{P}(\text{DAS}) + \mathbb{P}(\text{DAR}|\text{DOS}) \cdot \mathbb{P}(\text{DOS})} \\ &= \frac{(\frac{2}{3})(\frac{4}{7})}{(\frac{2}{3})(\frac{4}{7}) + (\frac{1}{4})(\frac{3}{7})} = \frac{32}{41},\end{aligned}$$

which is precisely the probability that a dash has been sent if a dash is received. Next,

$$\begin{aligned}\mathbb{P}(\text{DOS}|\text{DOR}) &= \frac{\mathbb{P}(\text{DOR}|\text{DOS})\cdot\mathbb{P}(\text{DOS})}{\mathbb{P}(\text{DOR}|\text{DOS})\cdot\mathbb{P}(\text{DOS})+\mathbb{P}(\text{DOR}|\text{DAS})\cdot\mathbb{P}(\text{DAS})}\\ &= \frac{(\frac{3}{4})(\frac{3}{7})}{(\frac{3}{4})(\frac{3}{7})+(\frac{1}{3})(\frac{4}{7})} = \frac{27}{43},\end{aligned}$$

which is the probability that a a dot has been sent if a dot is received.

(b) We need yet another probability with DOR condition,

$$\begin{aligned}\mathbb{P}(\text{DAS}|\text{DOR}) &= \frac{\mathbb{P}(\text{DOR}|\text{DAS})\cdot\mathbb{P}(\text{DAS})}{\mathbb{P}(\text{DOR}|\text{DAS})\cdot\mathbb{P}(\text{DAS})+\mathbb{P}(\text{DOR}|\text{DOS})\cdot\mathbb{P}(\text{DOS})}\\ &= \frac{(\frac{1}{3})(\frac{4}{7})}{(\frac{1}{3})(\frac{4}{7})+(\frac{3}{4})(\frac{3}{7})} = \frac{16}{43}.\end{aligned}$$

Independence greatly simplifies calculations. We have DOR two times, so the possible signals sent with corresponding probabilities are,

$$\begin{aligned}\mathbb{P}(\text{DOS}|\text{DOR})\cdot\mathbb{P}(\text{DOS}|\text{DOR}) &= \frac{27}{43}\cdot\frac{27}{43},\\ \mathbb{P}(\text{DOS}|\text{DOR})\cdot\mathbb{P}(\text{DAS}|\text{DOR}) &= \frac{27}{43}\cdot\frac{16}{43},\\ \mathbb{P}(\text{DAS}|\text{DOR})\cdot\mathbb{P}(\text{DOS}|\text{DOR}) &= \frac{16}{43}\cdot\frac{27}{43},\\ \mathbb{P}(\text{DAS}|\text{DOR})\cdot\mathbb{P}(\text{DAS}|\text{DOR}) &= \frac{16}{43}\cdot\frac{16}{43},\end{aligned}$$

which is the final distribution we are looking for.

Exercise 1.7. Standardized tests provide an interesting application of probability theory. Suppose first that a test consists of 20 multiple-choice questions, each with 4 possible answers. If the student guesses on each question, then the taking of the exam can be modeled as a sequence of 20 independent events. Find the probability that the student gets at least 10 questions correct, given that she is guessing.

2 Random variables

Exercise 2.1. Consider a random variable X with the following probability density function and $c < \infty$ being some constant:

$$f_X(x) = \begin{cases} cx(1-x), & \text{if } x \in (0,1)\\ 0, & \text{otherwise.}\end{cases}$$

(a) Determine the value of the parameter c that makes $f_X(x)$ a pdf. Derive the cumulative density function (cdf) of X.

(b) Derive mean, variance, skewness, and kurtosis of X.

(c) Consider another random variable $Y = X^3$, and derive its pdf.

(d) Derive mean, variance, skewness, and kurtosis of Y.

(e) For $g(x) = x^3$ compare $\mathbb{E}[g(X)]$ and $g(\mathbb{E}[X])$. What about $z(x) = x^{\frac{1}{3}}$?

(f) Determine the value of c that makes $f(x) = c \cdot e^{-|x|}$, $-\infty \le x \le \infty$ a pdf.

Solution.

(a) By the definition of the pdf,

$$\int_0^1 x(1-x)dx = \int_0^1 (x-x^2)dx = \int_0^1 xdx - \int_0^1 x^2dx$$
$$= \frac{x^2}{2}\Big|_0^1 - \frac{x^3}{3}\Big|_0^1 = \frac{1}{2} - 0 - \frac{1}{3} + 0 = \frac{1}{6}.$$

Thus, we need $c = 6$ for the pdf to integrate to 1 over the entire support.

By the definition of the cdf,

$$F_X(x) = \int_0^x 6x(1-x)dx$$
$$= 6\left(\frac{x^2}{2}\Big|_0^x - \frac{x^3}{3}\Big|_0^x\right) = 6\left(\frac{3x^2-2x^3}{6}\right) = 3x^2 - 2x^3,$$

so that

$$F_X(x) = \begin{cases} 1, & \text{if } x \geq 1, \\ 3x^2 - 2x^3, & \text{if } x \in (0,1), \\ 0, & \text{if } x \leq 0. \end{cases}$$

(b) Again by the definition, we derive

- mean: $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)dx = \int_0^1 6x^2(1-x)dx = 6\left(\frac{x^3}{3}\Big|_0^1 - \frac{x^4}{4}\Big|_0^1\right) = 6\left(\frac{1}{3} - \frac{1}{4}\right) = 0.5,$
- variance[2]: $\text{var}[X] = \mathbb{E}[(X - \mathrm{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = 0.3 - (0.5)^2 = 0.05$, because the second moment is
$$\mathbb{E}[X^2] = \int_0^1 6x^3(1-x)dx = 6\left(\frac{x^4}{4}\Big|_0^1 - \frac{x^5}{5}\Big|_0^1\right) = 6\left(\frac{1}{4} - \frac{1}{5}\right) = 0.3,$$
- skewness, κ_3, which is a rescaled version of the third *centered* moment:
$$\mathbb{E}[(X - \mathbb{E}[X])^3] = \int_0^1 (x - \mathbb{E}[x])^3 f_X(x)dx = \int_0^1 (x-0.5)^3 6x(1-x)dx$$
$$= 6\int_0^1 (x^3 - 1.5x^2 + 0.75x - 0.125)x(1-x)dx$$
$$= 6\left(-\frac{x^6}{6} + \frac{x^5}{2} - \frac{9x^4}{16} + \frac{7x^3}{24} - \frac{x^2}{16}\right)\Bigg|_0^1 = 0,$$

  so that the skewness is
$$\kappa_3 = \frac{\mathbb{E}[(X - \mathbb{E}[X])^3]}{(\mathbb{E}[(X - \mathbb{E}[X])^2])^{\frac{3}{2}}} = 0.$$
- kurtosis, κ_4, which is a rescaled version of the *forth* moment,
$$\mathbb{E}[(X - \mathbb{E}[X])^4] = \int_0^1 (x - \mathbb{E}[x])^4 f_X(x)dx = \int_0^1 (x-0.5)^4 6x(1-x)dx$$
$$= 6\left(-\frac{x^7}{7} + \frac{x^6}{2} - \frac{7x^5}{10} + \frac{x^4}{2} - \frac{3x^3}{16} + \frac{x^2}{32}\right)\Bigg|_0^1 \approx 0.005,$$

  so that the kurtosis is
$$\kappa_4 = \frac{\mathbb{E}[(X - \mathbb{E}[X])^4]}{(\mathbb{E}[(X - \mathbb{E}[X])^2])^2} \approx 2.14.$$

  Note that the skewness is equal to zero, the fact that follows from the symmetry of the distribution (important case is a normal distribution).

[2]Note that you can also proceed by the definition $\text{var}[X] = \int_0^1 (x - \mathbb{E}[x])^2 f_X(x)dx$.

If skewness is positive, the distribution is positively skewed or skewed right, meaning that the right tail of the distribution is longer than the left. If skewness is negative, the distribution is negatively skewed or skewed left, meaning that the left tail is longer. If skewness is zero, the distribution is symmetrical.

Kurtosis is a measure of the combined sizes of the two tails. If the kurtosis is greater than 3, then the distribution has heavier tails than a normal distribution. If the kurtosis is less than 3, then the distribution has lighter tails than a normal distribution. For a normal distribution we have kurtosis equal to 3. Note that kurtosis is always greater or equal to 0.

(c) Let us proceed by the definition of a cdf,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \le y) = \mathbb{P}(X^3 \le y) = \mathbb{P}(X \le y^{1/3}) \\ &= \int_0^{y^{1/3}} 6x(1-x)dx = (3x^2 - 2x^3)\Big|_0^{y^{1/3}} = 3y^{2/3} - 2y. \end{aligned}$$

Note that the support does not change, since $g(0) = 0$ and $g(1) = 1$. The full answer would be:

$$F_Y(y) = \begin{cases} 1, & \text{if } y \ge 1, \\ 3y^{2/3} - 2y, & \text{if } y \in (0,1), \\ 0, & \text{if } y \le 0. \end{cases}$$

The first equality is the definition; the second equality uses the transformation $g(x) = x^3$; the third inequality applies the inverse $g^{-1}(x) = x^{1/3}$ which is increasing on the support of X $(0,1)$ (and thus, preserves the order); the fourth inequality is again a definition.

Using the definition, we can also calculate the pdf,

$$f_Y(y) = \frac{dF_Y(y)}{dy} = 2y^{-1/3} - 2,$$

for $y \in (0,1)$ and 0 otherwise.

(d) Again by the definition,

- mean: $\mathbb{E}[Y] = \int_{-\infty}^{\infty} y f_Y(y)dy = 0.2$,
- variance: $\operatorname{var}[Y] = \int_{-\infty}^{\infty} (y - \mathbb{E}[y])^2 f_Y(y)dy = \dfrac{13}{300} \approx 0.04$,
- skewness: $\kappa_3[Y] = \dfrac{\mathrm{E}[(X - \mathrm{E}[X])^3]}{\mathrm{E}[(X - \mathrm{E}[X])^2]^{\frac{3}{2}}} = \dfrac{\frac{63}{5500}}{(\frac{13}{300})^{\frac{3}{2}}} \approx 1.27$,
- kurtosis: $\kappa_4[Y] = \dfrac{\mathrm{E}[(X - \mathrm{E}[X])^4]}{\mathrm{E}[(X - \mathrm{E}[X])^2]^2} = \dfrac{\frac{713}{96250}}{(\frac{13}{300})^2} \approx 3.95$.

(e) Here, the following result is useful:

Theorem 2.2 (Jensen's inequality)**.** *For any random variable X, if $g(x)$ is a convex function, then*

$$g(\mathbb{E}[X]) \le \mathbb{E}[g(X)].$$

If $g(x)$ is a concave function, the inequality is reversed.

Proof. We focus on the convex case. Let $a + bx$ be the tangent line to $g(x)$ at $x = \mathbb{E}[X]$. Since $g(x)$ is convex, $g(x) \ge a + bx$. Evaluating at $x = X$ and taking expectations, we find

$$\mathbb{E}[g(X)] \ge a + b\mathbb{E}[X] = g(\mathbb{E}[X])$$

as claimed. □

Now, we have

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx = \int_0^1 x^3 6x(1-x) dx = 0.2,$$
$$g(\mathbb{E}[X]) = g(0.5) = 0.5^3 = 0.125.$$

so $\mathbb{E}[g(X)] > g(\mathbb{E}[X])$ in line with Theorem 2.2.
For $z(x) = x^{1/3}$ we have

$$\mathbb{E}[z(X)] = \int_{-\infty}^{\infty} z(x) f_X(x) dx = \int_0^1 x^{1/3} 6x(1-x) dx \approx 0.77,$$
$$z(\mathbb{E}[X]) = z(0.5) = 0.5^{\frac{1}{3}} \approx 0.79,$$

so $\mathbb{E}[z(X)] < z(\mathbb{E}[X])$. It is, again, in line with Theorem 2.2 because $z(x)$ is concave.

(f) $\int_{-\infty}^{\infty} e^{-|x|} dx = \int_{-\infty}^{0} e^x dx + \int_0^{\infty} e^{-x} dx = 1 + 1 = 2$, so we need $c = \dfrac{1}{2}$ for pdf to integrate to 1 over the entire support. Non-negativity is obvious.